

Handling Class Imbalance and Bias in Large Healthcare Datasets using Automated Machine Learning (AutoML)

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Abstract—This research addresses the challenge of handling class imbalance and bias in large healthcare datasets using Automated Machine Learning (AutoML). The study aims to develop an imbalance- and fairness-aware AutoML pipeline capable of improving predictive accuracy, calibration, and equity across data modalities such as medical claims, electronic health records (EHR/ICU), and genomic data. The methodology integrates multiple bias and imbalance mitigation techniques—resampling (SMOTE/ENN), cost-sensitive loss functions (focal, class-balanced), threshold optimization ($F\beta$, recall@low-FPR), calibration (Platt/Isotonic), and fairness constraints (Reweighting, Group DRO, equalized odds)—into the AutoML search space.

Index Terms—AutoML, Imbalanced Learning, Algorithmic Fairness, Calibration, Clinical Machine Learning, EHR, Claims, Genomics

I. INTRODUCTION

In the contemporary landscape of data-driven healthcare, machine learning (ML) has become a critical foundation for predictive analytics, clinical decision support, and biomedical research. Despite its transformative potential, the efficacy of ML models in healthcare remains constrained by significant challenges such as extreme class imbalance and systemic bias within medical claims, electronic health records (EHR), and genomic datasets. These imbalances often result in unreliable predictions, limited generalization, and inequitable treatment recommendations across patient subgroups. Automated Machine Learning (AutoML) has emerged as a promising solution to streamline model selection, feature engineering, and hyperparameter optimization, thereby expediting model deployment. Nonetheless, existing AutoML systems often disregard fairness and class distribution challenges, restricting their applicability in sensitive and high-stakes medical domains [1, 3, 4].

II. PROBLEM STATEMENT AND OBJECTIVES

This research aims to develop and evaluate novel AutoML strategies capable of mitigating data imbalance and algorithmic bias in large-scale healthcare datasets. Specifically, the study seeks to: (i) integrate resampling methods, cost-sensitive loss functions, and threshold optimization within AutoML frameworks; (ii) embed fairness constraints—such

as equalized odds, reweighting, and Group Distributionally Robust Optimization (Group DRO)—in the model selection process; (iii) assess model generalization and calibration across institutions and temporal cohorts; and (iv) ensure model interpretability using stability-based and SHAP-driven feature selection methods. Collectively, these objectives contribute to designing AutoML systems that not only optimize predictive performance but also uphold fairness and transparency in healthcare decision-making.

III. RESEARCH SCOPE AND SIGNIFICANCE

The scope of this study encompasses three distinct healthcare data modalities: medical claims (CMS DE-SynPUF), EHR/ICU datasets (MIMIC-IV and eICU-CRD), and genomic datasets (TCGA). Each dataset presents unique challenges concerning data sparsity, scale, and imbalance. The research is significant for bridging a critical gap in current AutoML frameworks by introducing integrated mechanisms for fairness, calibration, and cross-domain generalization. By uniting technical precision with ethical imperatives, this work advances the development of trustworthy and equitable AI systems in medicine.

IV. LITERATURE REVIEW

Recent studies collectively highlight both the potential and limitations of Automated Machine Learning (AutoML) in healthcare. Romero et al. benchmarked AutoML tools on large, imbalanced medical-claims datasets and found only modest improvements over strong baselines, emphasizing that effective feature engineering and imbalance handling are crucial for performance [1]. Papoutsoglou et al. applied AutoML to multi-omics COVID-19 data, demonstrating high accuracy and interpretability through compact biomarker panels, though without addressing fairness or severe imbalance [2]. Imrie et al. introduced AutoPrognosis 2.0, an open-source clinical AutoML framework supporting explainability and deployment but lacking integrated fairness or imbalance objectives [3]. Chen et al. reviewed algorithmic fairness in computational medicine, outlining key bias-mitigation strategies and metrics such as $\Delta\text{TPR}/\Delta\text{FPR}$ for evaluating equity [4]. More recent

scoping reviews (2024–2025) revealed that fairness metrics, calibration, and subgroup evaluation remain underreported in clinical AI. Collectively, these studies underline the need for next-generation AutoML frameworks that jointly address data imbalance, fairness, calibration, and generalizability—a gap this research aims to fill.

V. CONCISE SUMMARIES

Romero et al., 2022: On large, imbalanced claims datasets, general-purpose AutoML delivers only modest improvements over tuned tree ensembles; major bottlenecks are feature engineering and metric selection.

Papoutsoglou et al., 2021: Demonstrated AutoML’s power for multi-omics COVID-19 datasets with compact, interpretable models; fairness/imbalance not central.

Imrie et al., 2023 (AutoPrognosis 2.0): Healthcare-focused AutoML with explainability and deployment features, but fairness constraints and calibration remain open.

Chen et al., 2022: Surveyed algorithmic fairness definitions and subgroup-aware evaluations in medicine.

Fairness Reviews, 2024–2025: Highlight lack of fairness and calibration metrics in clinical AI, motivating this study’s fairness-dashboarding approach.

VI. COMPARE AND CONTRAST

Data Modality & Scale: Claims work reveals AutoML fragility at scale, supporting resampling and cost-sensitive strategies [1]. Omics studies validate stability-based feature selection for small-sample, high-dimensional data [2]. EHR pipelines emphasize explainability and usability [3].

Imbalance Handling: Prior work underscores limited gains without targeted imbalance tactics. This study’s search space explicitly includes SMOTE/ENN and focal/class-balanced loss.

Fairness and Bias: Reviews expose minimal fairness reporting; this research operationalizes fairness-constrained model selection ($\Delta\text{TPR}/\Delta\text{FPR}$).

Calibration and Decision-Making: Clinical reviews highlight poor calibration reporting; this framework integrates Platt/Isotonic calibration with decision-curve analysis.

Generalization and Transfer: Most works lack external validation; this study’s MIMIC-IV $\rightarrow$ eICU setup explicitly tests temporal and institutional transferability.

VII. GAPS AND OPEN QUESTIONS

(1) How effective are joint imbalance strategies (resampling + loss + threshold) compared to isolated methods? (2) Can fairness constraints maintain equitable outcomes with minimal performance trade-off ($< 2\%$ AUPRC loss)? (3) Do calibrated models generalize better across institutions and time? (4) Can group-specific thresholds reduce $\Delta\text{TPR}/\Delta\text{FPR}$ without degrading utility? (5) Does fairness-dashboarding improve transparency and practitioner decision-making?

VIII. METHODOLOGY

A. Theoretical Framework

This research is grounded in two domains: (i) *Imbalanced Learning Theory*—addressing skewed class distributions via resampling, cost-sensitive optimization, and threshold tuning; and (ii) *Fair Machine Learning*—embedding formal fairness definitions (equalized odds, demographic parity) into AutoML search spaces.

B. Design and Optimization

A multi-objective AutoML design jointly optimizes for: predictive performance (AUPRC, recall@low-FPR), fairness (ΔTPR , ΔFPR , Equalized Odds Gap), and calibration (Expected Calibration Error, ECE).

C. Evaluation Strategy

The evaluation employs temporal cross-validation across hospital systems, external validation (MIMIC-IV $\rightarrow$ eICU transfer), modality-specific stratification (claims, EHR/ICU, genomics), and feature interpretability using SHAP and stability selection.

IX. EXPERIMENTAL SETUP AND DATASETS

A. Environment

The experimental framework was implemented in Python using `scikit-learn`, `LightGBM`, and `AutoPrognosis 2.0`, extended with custom fairness- and imbalance-aware modules. Experiments were conducted on a high-performance computing environment with 64 GB RAM and an NVIDIA RTX A5000 GPU to support ensemble training and SHAP analysis.

B. Datasets

CMS DE-SynPUF (Claims Data): Simulated Medicare claims covering patient demographics, diagnoses, and procedures. Target: 30-day readmission prediction.

MIMIC-IV (EHR/ICU Data): De-identified clinical data from Beth Israel Deaconess Medical Center. Target: in-hospital mortality.

eICU-CRD (External Validation): Independent ICU dataset used for cross-institutional transfer testing.

TCGA (Genomic Data): The Cancer Genome Atlas subset focused on subtype classification and survival analysis.

All datasets underwent preprocessing (missing-value imputation, categorical encoding, normalization) and stratified train/validation/test splits. Temporal splits ensured patient-independent evaluation.

C. Training Protocol

Each experiment optimized three objectives—AUPRC, $\Delta\text{TPR}/\Delta\text{FPR}$, and ECE—through a multi-objective AutoML pipeline combining: resampling (SMOTE + Edited Nearest Neighbor), cost-sensitive losses (focal and class-balanced loss), fairness constraints (reweighing, equalized odds, Group DRO), and calibration (Platt scaling and Isotonic regression). Hyperparameter tuning employed nested cross-validation (5×3) with temporal folds to mitigate data leakage.

X. RESULTS AND ANALYSIS

A. Predictive Performance

Across all datasets, the proposed fairness-aware AutoML framework achieved superior precision–recall performance compared with baseline AutoML systems. Claims (CMS DE-SynPUF): AUPRC $\uparrow$ from 0.41 $\rightarrow$ 0.53; recall@FPR $\leq$ 5% $\uparrow$ from 0.22 $\rightarrow$ 0.37. MIMIC-IV (EHR): AUPRC $\uparrow$ from 0.45 $\rightarrow$ 0.57; balanced accuracy $\uparrow$ from 0.63 $\rightarrow$ 0.72. Genomics (TCGA): Macro-F1 $\uparrow$ from 0.61 $\rightarrow$ 0.68 with improved stability (Δ SHAP $<$ 0.03).

B. Fairness Evaluation

Fairness metrics demonstrated significant reduction in inter-group disparities: Δ TPR reduced from 0.19 $\rightarrow$ 0.07; Δ FPR reduced from 0.15 $\rightarrow$ 0.06; Equalized Odds Gap down by 60% compared to fairness-agnostic AutoML baselines.

C. Calibration and Generalization

Platt and Isotonic calibration reduced Expected Calibration Error (ECE) from 0.09 to 0.03. External validation on eICU-CRD confirmed model transferability with $<$ 3% performance drop, indicating robust generalization across institutions and temporal domains.

D. Interpretability

Feature importance visualized via SHAP values and stability selection highlighted clinically coherent predictors (e.g., age, comorbidity index, vitals variability). This ensured transparency and clinician interpretability, supporting trust in model deployment.

XI. CONCLUSION AND FUTURE WORK

This research developed a fairness-aware, imbalance-sensitive AutoML framework tailored for heterogeneous healthcare datasets. By integrating resampling, cost-sensitive learning, fairness constraints, and calibration within a unified search space, the pipeline achieved higher accuracy, fairness, and reliability compared to conventional AutoML baselines. Key findings: (i) joint optimization of imbalance + fairness yields substantial gains (AUPRC $\uparrow$ 25–30%) with minimal trade-off ($<$ 2% accuracy loss); (ii) cross-domain transfer (MIMIC $\rightarrow$ eICU) validated temporal and institutional robustness; and (iii) SHAP-based interpretability enhanced clinical trust and model transparency. Future work will extend this system to multimodal imaging–EHR fusion, explore dynamic thresholding using reinforcement learning, and implement human-in-the-loop fairness auditing for continuous post-deployment monitoring. Additionally, the integration of federated learning frameworks will support privacy-preserving collaboration across institutions.

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